

Week 04

Protocols

Web Data Science

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Tuesday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 00:65 → Notebook lecture & exercises
- 00:65 – 00:70 → Thursday show & tell
- 00:70 – 00:75 → Next Class

Daily Note questions

- What is the different between data mining and web scraping?
 - Data mining is a larger set of methods for all kinds of data (biological, astrophysical, etc.) and web scraping is focused on retrieval and analysis of data from web sources
- What is your favorite podcast?
 - Endless Thread was a fun podcast about Reddit, The War on Cars if you want to get political

Module Assignment 01

- **Objective:** Write up a research proposal for analyzing the legislative activity of the Colorado Legislature.
- **Deliverable:**
Minimum five-page slide deck submitted as a PDF to Canvas by **Monday, September 23**
 - **Motivation:** What is the question or problem? Why is it important? Why could this data answer it?
 - **Data source:** What is the specific construct to measure? Screenshot a coleg.gov web page and circle it
 - **Methods:** What data will you need to collect to address your question? What will you do with the data?
 - **Expected results:** What kinds of analyses or visualizations will you do with the data to answer the RQ?
 - **Implications & Limitations:** What should Coloradans do if your RQ is right? What are limits to your analysis?
- **Evaluation:** More details in each will get higher scores. Screenshots, sketched visualizations, linking to scholarly or journalistic coverage, admitting limitations
- See Week 3 slides for more details and examples!

Web protocols

- Transmission Control and Internet Protocols (TCP/IP) enables transport/delivery of data
- Domain Name System (DNS) protocol converts a domain into an IP address
- Hypertext Transfer Protocol (HTTP) in an application layer for delivering hypertext documents
- Uniform Resource Locator (URL) identifies an information location and standard for access

TCP/IP

- Every computer on a public network has an IP address
 - Important servers (Google, etc.) have static IP addresses, while your laptop has a dynamic IP address
- IPv4 addresses formatted as 4 “octets”
 - [0-255].[0-255].[0-255].[0-255] → 2^{32} possible values (4e9 possible values)
- IPv6 addresses formatted as 8 hexadecimal numbers
 - FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF → $3.4e38$ possible values

DNS

```
subdomain.domain.top-level-domain
```

- IANA controls the availability of top level domains like `com`, `org`, `edu`, `us`, `social`, `io`
- You can register new domains with various services (GoDaddy, NameCheap, Dreamhost)
- Subdomains are controlled by the domain owner and don't need to be registered
 - `en.wikipedia.org`, `es.wikipedia.org`, `m.wikipedia.org`
- DNS is basically a “phone book” translates a domain name to an IP address where the server lives
- Use the `socket` library's `gethostbyname` to lookup IP addresses from a domain name

URLs

```
scheme : // userinfo @ host : port / path ? query # fragment
```

- `scheme` – description of application protocol to use (http, https, ftp, file, mailto, doi, ...)
- `userinfo` – (uncommon) username to log into a protected host
- `host` – a registered name (to be resolved by DNS) or an IP address locating the host
- `path` – a path to the resource being requested on the host
- `query` – (optional) assigns values to parameters to be parsed by the web server
- `fragment` – (optional) points to a secondary source, commonly section headings in a page

- You may think you want to write regular expressions to parse URLs → NO!
- Use a library following formal specifications like `urlparse` from [`urllib.parse`](#)

Next class

- Guest lecture from Andrew Brandt
 - Founder of Elect More Hackers
 - Former data journalist and current principal researcher at Sophos
- Daily note: <https://bit.ly/info4871f24note>
- Today's number is...

Thursday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 00:70 → Guest lecture
- 00:70 – 00:75 → Daily note

Daily Note questions